

Vrije Universiteit Amsterdam



Bachelor Thesis

Testing Hybrid Detection Approaches for Post-Earthquake Building Collapse Detection

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1. Abstract

Rapid and accurate detection of collapsed buildings following earthquakes is critical for effective emergency response. This thesis investigates a hybrid framework combining machine learning and physical attribute analysis for post-earthquake collapse detection using satellite imagery. The approach integrates a UNet++ segmentation model trained on pre- and post-disaster images with a rule-based shadow analysis module that leverages sun-angle metadata to predict collapse via shadow reduction. These two components are fused through a soft-voting ensemble to produce a per-pixel collapse probability map. The system is tested on high-resolution imagery of Antakya, Turkey, using Humanitarian OpenStreetMap annotations for validation. Results show that while the UNet++ model struggles with generalization and class imbalance, the ensemble improves recall and interpretability by incorporating physically grounded shadow cues. This work highlights the potential and limitations of hybrid AI systems in disaster response and offers a practical foundation for future research in rapid damage assessment.

2. Introduction

2.1. Motivation: Earthquakes and Urban Vulnerability

Natural disasters have plagued human development for what seems like forever. Nowhere is this more evident than in the case of earthquakes. From biblical references to modern-day Haiti, tectonic movements cause mass destruction and a death toll often in the tens of thousands¹⁻³. While the frequency of other disasters varies and can sometimes be altered or reduced by human intervention⁴, earthquakes pose a constant and immutable threat⁵. This danger is exacerbated by increasing population density, unregulated urban sprawl, lax building standards, and aging infrastructure—conditions commonly found in seismically active regions^{6, 7}.

Most notable among the dangers posed by earthquakes is structural collapse⁸. Buildings—especially those not designed to withstand significant seismic forces—crumple to the ground, trapping or injuring those inside⁹⁻¹¹. Moreover, the collapse of homes, hospitals, schools, and infrastructure leads to prolonged humanitarian crises and massive logistical challenges for emergency response teams^{12, 13}.

One of the defining characteristics of post-earthquake rescue operations is their urgency. The first 72 to 120 hours following a major seismic event are considered the “golden period” for search and rescue^{14, 15}. Beyond this window, the likelihood of finding survivors beneath the rubble diminishes sharply. As a result, time becomes a critical and limiting factor: emergency responders must act quickly and decisively, often with minimal information and in extremely chaotic environments.

Given these constraints, the ability to rapidly identify collapsed buildings is not merely helpful—it is essential. Search-and-rescue teams need to prioritize their limited resources toward areas with the highest probability of entrapment. Misallocating these resources to undamaged or less-affected structures can delay life-saving interventions and lower the overall success rate of rescue missions¹⁶. Thus, developing systems that can reliably detect building collapses immediately after an earthquake is one of the most important technological challenges in the disaster response field today¹⁷.

2.2. Current detection possibilities and challenges

The near-constant monitoring of our planet by satellites provides us with a promising tool to tackle this challenge. Modern satellite apparatus such as MAXAR are able to capture high-resolution, multispectral images of affected areas within hours of a disaster. These images, when compared to pre-disaster footage of the same location, allow us to observe direct visual changes caused by the earthquake. For example, in the city of Antakya—one of the hardest-hit areas during the 2023 Turkey-Syria earthquake—we can clearly see the dense and closely knit urban development in the pre-event images, and vast swaths of debris where buildings once stood in the post-event images. As exemplified below.



Figure 1 - Example of difference between pre- and post-earthquake images.

However, manual interpretation of these images is time-consuming and labor-intensive. This is where machine learning comes into play. Over the past decade, these algorithms—particularly convolutional neural networks (CNNs)—have shown promise in automatic classification, and segmentation of various patterns (especially in the medical field)^{18, 19}. These characteristics exemplified in architectures such as UNet and its successor UNet++ can also be harnessed in the context of damage/collapse detection²⁰.

Despite these advancements, algorithms like UNet still suffer from several limitations -they often fail to generalize well when there are significant variations in lighting, building styles, or occlusions not represented in the training data, leading to a high false-positive rate²¹. This remains a major concern, as incorrect labeling could potentially lead to the misallocation of critical rescue resources²².

2.3. The promise of hybrid strategies

In response to these challenges, hybrid approaches—where machine learning algorithms are ensembled with physically grounded features—offer hope for alleviating the weaknesses of purely model-driven methods²³. One readily available physical feature is shadow. Due to the direct correlation between shadow length and building height (in combination with satellite metadata), one can infer a building’s height from its shadow—or lack thereof. By comparing shadows in pre- and post-event images, a significant reduction in shadow length likely indicates a collapse. This physically interpretable signal provides an additional validation mechanism for the “black box” predictions produced by machine learning models.

2.4. Research Contribution

This thesis introduces a hybrid framework that fuses UNet++-based semantic segmentation with a shadow-analysis module for building collapse detection in satellite imagery. The goal is to combine the adaptability of deep learning with the physical interpretability of shadow loss.

The system operates in two stages:

1. **UNet++** processes paired pre- and post-event images to predict collapse likelihood at the pixel level.
2. **Shadow analysis** computes shadow-loss ratios using sun-angle metadata and image features to further strengthen and balance-out predictions.

By averaging the outputs of both systems through a soft-voting ensemble, the final model produces a per-pixel collapse probability map that is aimed to be more precise and interpretable than predictions from either method alone.

The framework is evaluated in a case study of Antakya, Hatay Province, Turkey—a city devastated by the 2023 earthquake. Collapse data derived from Humanitarian OpenStreetMap Team (HOTOSM) annotations are used for validation. Special emphasis is placed on the model’s ability to reduce false positives while maintaining high recall—a key requirement for any system intended for operational deployment.

Ultimately, this research aims to offer a practical, scalable solution, uncovering the limitations of systems used to develop the software as well as in the algorithms/approaches themselves. And in doing so, contribute to the growing field of AI for humanitarian applications and address one of the most pressing needs in earthquake response: identifying where help is needed most.

3. Related Work

3.1. UNet and UNet++ Architectures

The UNet architecture, originally introduced by Ronneberger et al. (2015)²⁷, was designed for biomedical image segmentation but has since been widely adopted in various remote sensing applications due to its encoder-decoder structure and skip connections. UNet++ (Zhou et al., 2018)¹⁴ extends the original UNet with nested and dense skip pathways to reduce the semantic gap between encoder and decoder features. These improvements have demonstrated increased accuracy in fine-grained segmentation tasks, particularly where training data are limited or classes are imbalanced.

UNet++ has been explored for post-disaster damage mapping, with studies like that of Nam Vu¹⁶ demonstrating its advantages over traditional UNet in identifying collapsed buildings using satellite imagery. In his case study on the 2023 Turkey–Syria earthquakes, Vu showed that while UNet++ achieved slightly higher F1 scores than UNet, both models suffered performance drops on real-world test data, highlighting issues of generalizability and performance on lower-quality/higher complexity cases.

3.2. Building Damage Detection with Deep Learning

Several deep learning models have been developed to improve damage detection accuracy and reduce false positives. BD-SKUNet (Ahmadi et al., 2024) is one such example, incorporating selective kernel modules into a two-stage UNet-like architecture. The model outperformed baselines on both the xBD dataset and new data from the Bam earthquake in Iran, indicating strong generalization capabilities. Similarly, ECADS-CNN (Zhao et al., 2024) introduced a U-Net for segmentation followed by a Siamese network for classification, showing promising results in segmenting and classifying building damage across multiple disaster types.

Other studies, such as Bai et al. (2018), applied standard U-Net models to damage mapping following the 2011 Tohoku earthquake and tsunami. Their model successfully categorized building states into “washed away,” “collapsed,” and “survived”.

3.3. Challenges with xBD and Ensemble Approaches

The xBD dataset, used in many of these studies, presents both opportunities and challenges. While it is one of the most comprehensive open datasets for disaster response, issues such as inconsistent annotation quality and class imbalance persist. Su et al. (2020) addressed these challenges by proposing a range of solutions, including GAN-based pre-disaster image synthesis, metric harmonization between pixel-level and object-level evaluation, and, most importantly for our work, oversampling of damaged buildings. They also introduced an ensemble strategy that combines predictions from semantic segmentation models with physically derived features.

3.4. Shadow-Based Methods for Structural Change Detection

Beyond pure deep learning models, some researchers have explored using shadows as a proxy for structural height and change. Terletzky and Ramsey (2014) introduced a semi-automated image differencing method to identify animals based on changes in shadow locations between paired aerial images. While not directly applied to structural collapse, their method established a precedent for using shadow geometry in change detection.

In the context of earthquake damage, this study builds on such ideas by incorporating a shadow-based validation mechanism. Shadows, when measured pre- and post-disaster using sun angle metadata, can provide a reliable estimate of changes in building height, serving as a verification mechanism for deep learning predictions.

3.5. Few-Shot Learning and Data Scarcity Solutions

To address limitations in data availability and imbalance, Koukouraki et al. (2022) explored few-shot learning techniques for urban damage detection. Their use of Prototypical Networks showed strong performance even with limited training examples, especially in detecting rare classes like collapsed buildings. Oversampling and cost-sensitive training were also evaluated, with the few-shot approach ultimately delivering the best balance across all classes.

3.6. Use of shadows for detection of earthquake-induced collapsed buildings in high-resolution satellite imagery

Tong et al. (2013) proposed a hybrid shadow-analysis approach for detecting earthquake-induced building collapses using high-resolution satellite imagery. By comparing the theoretical ground shadow polygon—calculated from pre-event building footprints and sun angles—with the actual shadow observed in post-earthquake images, the method can classify buildings as totally collapsed, partially collapsed, or uncollapsed. The approach demonstrated superior accuracy over existing techniques, particularly in dense urban settings where building shadows often overlap. This study gives us a framework for detecting and assigning shadows to buildings in an urban environment where often shadow distinction is difficult.

3.7. Convolution Neural Network for Building Damage Assessment: A Case Study of the 2023 Turkey–Syria Earthquakes

On top of the vast amount of contextual background knowledge, this paper's base is founded on Vu's bachelor thesis, where he implemented and compared UNET and its more advanced successor UNET++ trained on the xBD dataset, achieving promising results. Our thesis takes on many aspects from Nams work - the fundamental UNET++ architecture, dataset selection and unseen test case.

4. Data and Preprocessing

4.1. Dataset Overview

4.1.1 xBD

For the training of our Unet++ model we require vast amounts of pre- and post-disaster images along with labelings. For this purpose we chose the xBD dataset - a large openly available library of pre- and post-disaster images. These are stored in two tiers (tier1, tier3) and a curated “challenge” subset. Tier 3 contains a larger amount of data but of worse quality, while tier1 - the inverse. These two tiers raw amount to about 350GB once unzipped. In comparison the Challenge set is only about 67GB unpacked, with no description as to the contents of this subset, and how it differs from the two tiers. It is critical to understand the scale of these numbers, as processing (as described later in this work) is quite the challenge in itself.

On the whole, the dataset contains labeling of each building stored in a separate geojson file for each pre/post image pair. Each building polygon is labeled with one of the following categories:

Score	Label	Visual Description of the Structure
0	No damage	Undisturbed. No sign of water, structural damage, shingle damage, or burn marks.
1	Minor damage	Building partially burnt, water surrounding the structure, volcanic flow nearby, roof elements missing, or visible cracks.
2	Major damage	Partial wall or roof collapse, encroaching volcanic flow, or the structure is surrounded by water or mud.
3	Destroyed	Structure is scorched, completely collapsed, partially or completely covered with water or mud, or no longer present.

*Table 1 - Damage labelling of xBD dataset*²⁸

It is important to note that this dataset is far from perfect for a strictly earthquake-related work. It includes images from many other disaster types - hurricanes, floods, fires, etc. With only one strictly-earthquake disaster being included. Although the extra quantity is valuable it does raise some concerns as to the accuracy of algorithms trained on it. For example - in the case of floods it is often the case that buildings are not collapsed/heavily damaged, but the surrounding area and foliage is destroyed/significantly altered. This might impact the algorithm's ability to recognize rubble, due to the increase in surrounding entropy with no damage label assigned.

4.1.2 HOTOSM

For ground truth creation, as well as the functioning of our shadow model it is necessary to have the building footprints of the test region. Humanitarian OpenStreetMap Team (HOTOSM) provides us with two geoJSON files - one of the building footprints of collapsed buildings in the Turkey-Syria earthquake, and one of all buildings within Antakya. Since we are interested in the immediate city centre of Antakya it is necessary to crop these files to said area.

An important note is that the file of “all” buildings in Antakya does not include those collapsed in the earthquake, therefore a full geojson of all pre-earthquake buildings is achieved through the union of both files.

Once cropped and merged our final file provides a solid source of truth against which we can verify our results..

4.1.3 Case Study

This study focuses on the city of Antakya, located in Hatay Province, Turkey, one of the urban centers most severely impacted by the February 2023 Turkey–Syria earthquake sequence. The region was subjected to extreme seismic activity, with a maximum Mercalli intensity of XII recorded in the city center. As a result, Antakya experienced widespread structural collapse, making it an ideal testbed for evaluating post-disaster building damage detection algorithms. The urban landscape of Antakya is characterized by a dense, gridless pattern of low- to mid-rise residential and mixed-use buildings, many of which were constructed prior to modern seismic codes. The average building

height in the study area is estimated to be around 3 to 5 stories, or approximately 10 to 15 meters, with significant clustering that limits the visibility of building outlines and shadow separation in satellite imagery. This dense urban morphology, combined with varied roof materials and aging infrastructure, presents a complex visual environment for machine learning models.

For this case study, high-resolution pre- and post-event satellite imagery from MAXAR was used, alongside building footprint and damage annotations from the Humanitarian OpenStreetMap Team (HOTOSM). The ground truth data includes two layers: one with all building footprints and another indicating collapsed structures, both in vector format. These were rasterized and aligned with the satellite data to create pixel-level labels for model evaluation. Importantly, to preserve the integrity of the case study, no data from Antakya was included during model training—ensuring an unbiased test of the system’s generalization capabilities.

This region's urban typology, tight building spacing, and shadow occlusions posed significant challenges for both the deep learning and shadow-based algorithms. Evaluating model performance in such a complex real-world scenario provides valuable insights into the robustness and limitations of each method, particularly in post-earthquake urban environments where fast and accurate triage is essential.

4.1.4 MAXAR

As mentioned - for the purpose of testing, we need pre- and post-images of Antakya - a region not included in xBD. Thankfully the MAXAR Open Data Programme gives us access to a variety of pre- and post-disaster .tiff images. These are listed on their website (without metadata) and on their Github (with the accompanying metadata files). From this set we derive the antakya unseen test case. To match regions with that of Vu it was necessary to find the exact tile used there. Since the images are indexed by their Quad Code, one must take the coordinates of the desired region, in the case of Antakya city centre: 36.15724919432816 36.2050760135623. And convert them into a quad code using a shell command:

```
max-ard cell covers "POINT(36.15724919432816 36.2050760135623)"
```

Figure 2 - Shell script for converting coordinates to quad codes

This gives us a searchable index by which we find the correct .tiff and .json.

4.2 Universal Preprocessing

As we will later explain, many variations and combinations of data were attempted and tested. However, there is a set of preprocessing steps that are universally applied across all experiments of the UNET model. These consist of normalization, tiling, image stacking, and ground truth generation.

4.2.1. Tiling

Due to the large size of raw satellite images, they are divided into overlapping tiles of 256×256 pixels, with a stride of 224. This enables the model to focus on localized features and allows more efficient use of memory, as training can be done with smaller batches that fit within hardware constraints.

4.2.2. Normalization

All pixel values are normalized to the range $[0, 1]$. This standardizes the input scale and improves model stability and convergence.

4.2.3. Image Stacking

To support temporal comparison, each training sample is composed of a pre-disaster and a post-disaster image, stacked along the channel dimension. This results in a 6-channel input per tile (3 RGB channels from each image). Each of these tiles is saved as a .npz file containing the normalized 6-channel data.

4.2.4. Ground Truth Creation

The original labels contain multiple damage categories. For our purposes, these were simplified into three classes: background (0), uncollapsed buildings (1 — combining no damage and minor damage), and collapsed buildings (2 — combining major damage and destroyed). The labeled polygons are rasterized into a single-channel mask matching the tile size, saved as a .npz file alongside the image.

This process results in two .npz files per tile (image and mask), and when scaled up to hundreds of thousands of tiles, the file I/O overhead becomes significant. To mitigate this, tiles are grouped into batches of 4096 and saved as .tar archives. Each archive is roughly 8 GB, fitting comfortably in 16 GB of RAM for loading. This significantly reduces file access latency and improves training throughput. Future users should tune this batch size based on their system's RAM capacity.

Below we present a diagram of this process (see next page):

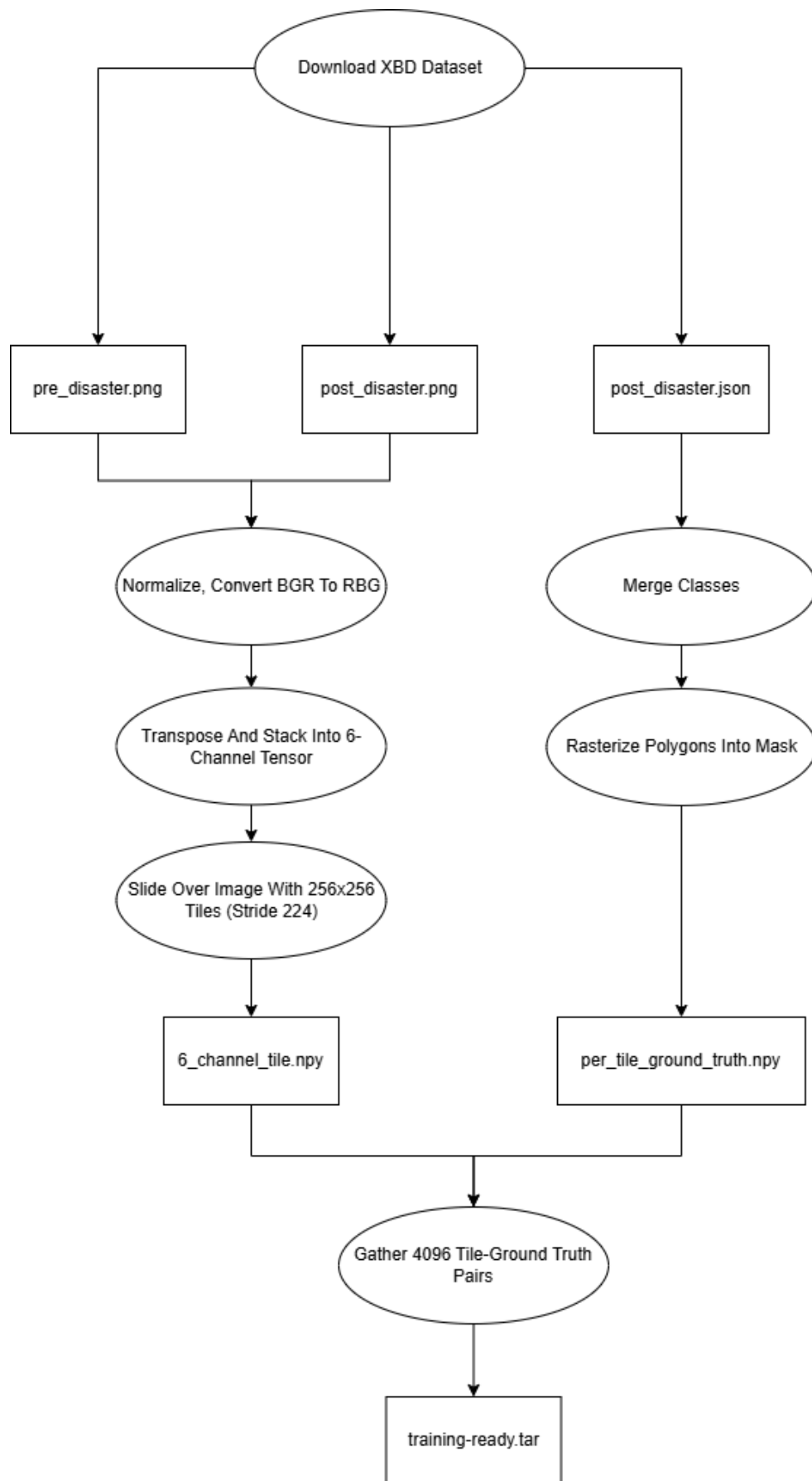


Figure 3 - Universal preprocessing diagram

4.3 Implications for Model Performance

It is important to note that these datasets are far from perfect for strictly earthquake-related work. For one - xBD includes images from many other disaster types - hurricanes, floods, fires, etc. With only one “pure” earthquake disaster being included. Although the extra quantity is valuable it does raise some concerns as to the accuracy of algorithms trained on it. For example - in the case of floods it is often the case that buildings are not collapsed/heavily damaged, but the surrounding area and foliage is destroyed/significantly altered. This might impact the algorithm's ability to recognize rubble, due to the increase in surrounding entropy with no damage label assigned.

In both the xBD and MAXAR datasets, one critical limitation that directly impacts shadow-based analysis is the inconsistency in sun-angle metadata between the pre- and post-disaster images. Ideally, for accurate shadow comparison, lighting conditions—including sun azimuth and elevation—should be closely matched between image pairs. However, in practice, satellite imagery is collected opportunistically rather than with controlled capture parameters. This results in significant discrepancies in illumination, which can alter the orientation, length, and visibility of shadows. Such variations introduce noise into the shadow-loss ratio calculations, potentially leading to incorrect inferences about structural collapse—particularly in borderline cases where the shadow difference is subtle.

Similarly, while the Humanitarian OpenStreetMap Team (HOTOSM) provides critical building footprint data and annotated collapse labels for the Antakya region, it lacks reliable information on building heights. Shadow projection relies on estimating the vertical dimension of structures, and in the absence of true height data, the model is forced to assume a uniform height value (e.g., 10 meters) across all buildings. This simplification reduces the spatial specificity of the shadow model and introduces a potential source of systematic error. For instance, mid-rise buildings may cast longer shadows than low-rise homes even if both remain standing, leading to underestimation or overestimation of damage based purely on shadow loss.

Together, these constraints highlight a fundamental challenge in applying physically grounded techniques like shadow analysis: they depend heavily on accurate metadata and contextual information, which is often incomplete or inconsistent in real-world disaster imagery. As a result, while hybrid approaches can improve reliability over black-box models alone, their accuracy remains bounded by the limitations of the underlying data.

5. Methodology

5.1 Overview of Framework

The hybrid system developed in this study consists of two main components: a UNet++ segmentation model and a shadow-based collapse validation module. The UNet++ network is used to produce initial collapse probability maps from pre- and post-event satellite image pairs. These predictions are then supported using shadow-loss ratios calculated from sun-angle metadata and building outlines.

A soft-voting ensemble strategy averages the outputs of both models, producing a final per-pixel collapse map. Let's now take a closer look at each element individually.

5.2 UNet++ Pipeline

5.2.1 Final Architecture and Configuration

The segmentation backbone employed in this framework is a UNet++ model, a nested encoder-decoder architecture that extends the conventional UNet design by introducing dense skip pathways. The model spans four downsampling levels, with the number of feature channels starting at 64 and doubling at each subsequent level (i.e., $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$). Each downsampling block includes two stacked 2D convolutional layers followed by ReLU activations (a "DoubleConv" block), and a 2D max pooling operation. The upsampling path mirrors this structure, using transposed convolutions for upsampling and integrating multi-scale features via skip connections.

Unlike traditional UNet, UNet++ incorporates nested skip connections between encoder and decoder blocks at different semantic levels. These connections form a series of intermediate decoder sub-networks that enable progressive feature fusion, significantly narrowing the semantic gap between encoder and decoder features. This nested design facilitates improved semantic consistency and spatial detail preservation, which is critical for high-resolution, pixel-level tasks like post-disaster building damage assessment.

The input to the model is a 6-channel tensor, formed by stacking co-registered pre- and post-disaster RGB images. This approach enables the network to jointly reason over temporal changes in the visual scene. The output is a single-channel probability map, where each pixel represents a predicted damage class (e.g., no damage, minor, major, destroyed), based on a sigmoid or softmax activation depending on the task formulation (binary vs. multiclass segmentation).

This architecture is heavily inspired by Nam Vu with minor tweaks such as the use of a 6 tensor input.

To better reflect the complexity of post-disaster imagery and to avoid the collapse of the model into background-only predictions, the classification task is formulated as a 3-class problem:

- Class 0: Background
- Class 1: Undamaged or minor damage
- Class 2: Major damage or collapsed structures

The model's final layer applies a softmax activation function to yield a per-pixel class probability across these three categories. During training, class imbalance is addressed using weighted cross-entropy loss, with weights of [0.0098, 0.1963, 1.0] empirically derived from their proportion of presence in the dataset:

Class	Label Index	Pixel Count	Proportion
Background	0	2,742,031,748	94.37%
No Damage	1	137,713,130	4.74%
Damage	2	26,907,795	0.93%

Table 2 - Table of class proportions in challenge set

Training was conducted exclusively on the Challenge subset of xBD, a smaller but more balanced and higher-quality portion of the dataset. This subset includes curated image pairs with improved annotation consistency and was the only viable option among the available data tiers that allowed the model to produce stable, multi-class predictions. The dataset was preprocessed into 256×256 pixel tiles with a stride of 224 and stored as .npy files grouped into .tar archives for efficient I/O during training.

5.2.2 Loss Function and Data subset iteration

A critical bottleneck in the development of an effective collapse detection model was identifying a viable combination of training data, loss function, and sampling strategy. Contrary to expectations based on existing literature, particularly the brief description provided by Nam Vu, initial attempts to replicate his performance using the full Tier 1 and Tier 3 datasets fell significantly short. This led to an extensive empirical investigation into how data curation, label balancing, and loss function formulation affect model learning.

5.2.2.1 Initial Experiments: Tier 1 + Tier 3, Binary Classification

The first round of experiments used the combined Tier 1 and Tier 3 datasets from the xBD repository. Given the extreme class imbalance—where collapsed structures make up only a fraction of the dataset—an aggressive oversampling strategy was applied. Training used a binary cross-entropy (BCE) plus Dice loss setup, and the model was trained on tiles with a 5x oversampling of those containing damage. Despite this, the model consistently collapsed to predicting only the background class.

Subsequent trials further increased positive sampling (10x) and introduced negative undersampling (0.5x). Loss functions were varied (e.g., Focal + Dice) to penalize underrepresented classes more heavily. These changes produced only marginal improvements, with Intersection-over-Union (IoU) remaining below 0.15. The model continued to either collapse to background-only predictions or output overly conservative segmentation masks.

5.2.2.2 Dataset Refinement: Tier 1 Only

Hoping that data quality might be a limiting factor, the training was restricted to the Tier 1 subset, which offers cleaner annotations. However, performance remained poor. Even with adjusted sampling ratios, the model again defaulted to predicting only background, reinforcing the conclusion that quantity alone could not overcome poor label signal.

5.2.2.3 Challenge Subset Transition

Eventually, the focus shifted to the “Challenge” subset of xBD—a smaller, curated collection with better-balanced and higher-quality data. Training on this dataset using the same BCE or CE losses produced modest improvements. However, results still skewed heavily toward the background class, and the model failed to consistently detect collapsed buildings.

At this stage, a review of Vu’s implementation revealed a subtle yet crucial detail: his model was trained exclusively on the Challenge subset and utilized class weighting, a point not clearly emphasized in his thesis text but observable in his shared code.

5.2.2.4 Switching to 3-Class Classification and Class Weights

Up to this point, all experiments had treated the task as a binary classification problem (collapsed vs. non-collapsed). A major conceptual shift occurred when the model architecture was adapted to a three-class formulation, distinguishing between:

- Class 0: Background
- Class 1: Undamaged/Minor Damage
- Class 2: Major Damage/Collapse

This reframing not only reflects a more realistic understanding of post-disaster imagery but also aligns better with the pixel-level label proportions.

To counteract the massive class imbalance within the Challenge dataset, weights were assigned based on inverse pixel frequency. As seen before in *Table 2*

introducing these weights led to a dramatic shift in model behavior. For the first time, the UNet++ architecture produced stable multi-class predictions that successfully distinguished between undamaged and damaged buildings. The best result yielded a Damage class F1-score of 0.35 and IoU of 0.15.

5.2.2.5 Fine-Tuning and Collapse Regression

Encouraged by these results, the weight for the “No Damage” class was varied in an attempt to improve class-specific recall. Increasing it beyond 0.25 caused the model to regress—once again collapsing to background-only or damage-only predictions. Adding Dice loss back into the loss function, while theoretically beneficial for boundary accuracy, destabilized the model, likely due to its over-sensitivity to class imbalance in this scenario.

These results highlight the razor-thin margins for success in highly imbalanced semantic segmentation tasks. Small adjustments in class weights or loss formulations led to dramatically different outcomes, ranging from complete collapse to partial success.

5.2.2.6 Key Takeaways

- Oversampling and focal losses alone cannot overcome poor signal in Tier 1 and 3 datasets.
- The Challenge subset, while smaller, provided the only viable training data source for meaningful results.
- The shift to 3-class classification with proper class weighting was a turning point. Weight tuning was highly sensitive; even modest increases in “No Damage” weight destabilized training.
- Nam Vu’s performance likely depended heavily on these fine-tuned weightings, even though they were not explicitly described.
- UNET model training in the form presented here is non-deterministic, with identical setups outputting drastically different results.

This iterative process, while time-consuming, underscored how delicate segmentation model training is when working with remote sensing data for disaster contexts. The quantity of data is no substitute for careful curation and deliberate loss balancing. Additionally, the decision to model “No Damage” as its own class—separate from both background and collapse—is not just helpful, but essential. Without it, the network fails to establish meaningful distinctions in structural state, even when other aspects of the pipeline are optimized.

A full table of the data is included in section 11.

5.3 Shadow-Based Algorithm

In parallel with the UNet++ model, a shadow-based module is introduced to estimate drastic changes in building height between pre- and post-disaster imagery. This method leverages the physical properties of cast shadows, which vary according to sun position and building elevation. As a rule-based approach, it does not require training and is therefore immediately applicable to the Antakya test case.

5.3.1 Theoretical Basis

The core assumption is that buildings which collapse lose height, resulting in reduced or missing shadows in the post-disaster image. Using the sun azimuth and elevation metadata, the expected direction and length of each building’s shadow is modeled. The method is based on the approach of Tong et al. (2013), where a shadow “search region” is projected for each building footprint, extending outward in the direction opposite to the sun’s azimuth.

Given the absence of per-building height information for Antakya, an average height of 10 meters is assumed for all buildings. The expected shadow length L (in meters) is calculated as:

$$L = \frac{H}{\tan(\theta)} \quad (1)$$

Where H is the assumed building height (10 meters), and θ is the sun elevation angle (in degrees). This formula allows for estimation of how long the building's shadow would be under given solar conditions.

This value is converted into pixels using the Ground Sample Distance (GSD) from the image metadata (e.g., 0.3 meters per pixel). Since sun elevation differs between image acquisitions, the shadow projection length is normalized by adjusting it using the ratio of the tangents of the respective sun elevations. This ensures consistent measurement across images with varying lighting conditions.

5.3.2 Shadow Segmentation

Both the pre- and post-event images are converted to grayscale, and shadows are identified via static thresholding. Morphological operations are then applied to clean the segmented masks—removing noise and filling small gaps. Within the projected shadow region of each building, the number of detected shadow pixels is counted in both the pre- and post-event images.

5.3.3 Shadow-Loss Ratio and Collapse Estimation

The degree of shadow reduction is quantified using the shadow-loss ratio (SLR), calculated as follows:

$$SLR = (Pre\text{-}event\ shadow\ pixels - Post\text{-}event\ shadow\ pixels) / Pre\text{-}event\ shadow\ pixels \quad (2)$$

This metric is interpreted as a proxy for collapse severity. A high SLR (e.g., > 0.8) indicates significant shadow loss and is suggestive of total collapse, while lower values imply partial or no structural failure. Although the relationship is modeled as linear in this implementation, it is acknowledged that this may oversimplify the true physical dynamics—a point further discussed in Section 9.

To maintain compatibility with the UNet++ output, the shadow algorithm generates a full-size probability map of shape (1, H, W). For each building footprint, the computed SLR is assigned as a uniform collapse probability to all pixels within the building. All other (non-building) pixels are set to zero.

5.3.4 Handling inconclusive cases

If shadows are unclear or metadata is insufficient, the algorithm deems the analysis for that building inconclusive. A building is excluded from shadow-based estimation and marked as uncertain if:

- Sun elevation exceeds 70 degrees
- Fewer than 10 pre-event shadow pixels are detected in the search region
- The building footprint is smaller than 20 square meters

- Metadata is missing or corrupted

Instead of leaving these pixels undefined, they are assigned a neutral probability of 0.5, indicating complete uncertainty (equal likelihood of collapse or no collapse). This allows the shadow-based map to remain complete and compatible for downstream ensembling.

When applied to the Antakya case study, the algorithm successfully produces a dense collapse probability map, with values reflecting the estimated severity of shadow reduction across the urban landscape.

5.3.5 Limitations of Shadow Analysis

During testing this algorithm accurately detected (with the exception of some dark foliage) and calculated shadow sizes. Capturing the differences in pre and post images. When checking accuracy however we see a rather underwhelming result. Upon inspection of all incorrect cases we can not edge cases which can “fool” the algorithm despite its accurate shadow detection.

Most notable of these is what for the purpose of this study we will call “shadow exposition”. When processing clustered buildings we can see a troubling behaviour - buildings which previously “blocked” a shadow once collapsed “reveal” the full shadow of the uncollapsed buildings around it. Leading to situations where the uncollapsed buildings “gain” height (due to an increase in shadow pixels in search area), these uncovered shadows are also often counted as the shadow of the collapsed building. Effectively masking the fact that the building is actually collapsed. Let’s see an example:



Figure 4 - Example of shadow extension after nearby building collapse in Antakya

In the pre-disaster part of the image (*Figure 4 - left*) we clearly see a row of buildings, each casting a narrow shadow due to being blocked by the next in line. In the post earthquake image however (*Figure 4 - right*) we see that some of the buildings in between are totally collapsed, leading the remaining buildings to cast longer shadows.

As shown - the extended shadow falls on the search area of the subsequent structures leading to it being falsely interpreted as the shadow of the collapsed building. The algorithm counts these shadow pixels indiscriminately, leading to collapsed buildings being overlooked due to lack in shadow pixel count (or even at times - an increase!)

Another difficulty with dense urban scenarios are simply - lack of shadow differences. Again due to nearby building cutoff, the pre images contain a very narrow shadow - with very little difference in size to its collapsed counterpart. This is exasperated by the fact that the rubble of the collapsed building itself throws a small but still significant shadow, meaning in some cases the difference is next to zero, as seen in *Figure 5*:



Figure 5 - Example of static shadow size

Potential fixes might be to account for nearby building density, subtracting or throttling shadow pixel counts if they are in the potential shadow area of other buildings. Another option would be to measure the entropy in the building footprint. Negating/countering the shadow prediction of a high difference in entropy is found. These intricacies however go beyond the initially predicted scope of this project and were therefore not tackled. This aspect of the work however is very promising for future research. Major improvements could be achieved if one were to account for these unique cases plaguing our current model.

5.4 Ensemble Strategy

Both the UNet++ model and the shadow-based algorithm independently produce a probability map of shape (1, H, W), where each pixel represents the likelihood of structural collapse. To combine these outputs, the maps are averaged pixel-wise, using simple arithmetic mean:

Ensemble probability = (UNet++ probability + Shadow-based probability) / 2

This strategy ensures that buildings confidently marked as collapsed by both models receive a high ensemble score, while disagreements between the models are reflected as intermediate probabilities. This averaging approach also smooths out model-specific noise, which is particularly valuable given the UNet++ model's non-determinism and the shadow algorithm's sensitivity to lighting and occlusion.

In buildings flagged as "uncertain" by the shadow module (e.g., due to missing metadata or poor sun angle), a neutral probability of 0.5 is assigned in the shadow map. As a result, the ensemble score in these regions falls midway between the UNet++ output and the neutral baseline, preserving partial confidence from the learned model without overstating certainty.

The final output of the ensemble is a continuous probability map, but it can optionally be thresholded into binary collapse predictions for evaluation or operational use. The threshold value can be tuned based on desired recall/precision trade-offs; in this project, thresholds were varied from 0.3 to 0.7 during evaluation to explore different performance regimes.

To ensure spatial consistency, post-processing masks out areas outside known building footprints (as defined by HOTOSM) in the final map. This avoids false positives in roads, vegetation, or other non-structural regions where UNet++ occasionally misfires.

In summary:

- Both maps (UNet++ and Shadow) are normalized to [0, 1] scale
- Non-building areas in the shadow map are assigned 0
- Uncertain buildings in the shadow map are assigned 0.5
- Final output is the average of the two maps, optionally thresholded
- Building footprints serve as a spatial mask for final predictions

This ensemble approach strikes a balance between machine-learned flexibility and rule-based stability, producing a collapse detection system that is more generalizable and interpretable than either method on its own.

6. Infrastructure

6.1 Initial Inherited Environment

In the early stages of this research, training was attempted with the use of Google Colab and Google Drive, following precedent set by earlier UNet++ implementations such as Nam Vu's. However, this project quickly exposed the scalability limitations of these platforms. The Challenge dataset alone occupies ~67 GB, while experiments using Tier 1 and Tier 3 datasets pushed total data volume to ~400 GB.

Google Drive, in particular, is incapable of performing I/O operations on large folders containing many individual files without timing out, making any system dependent on Drive-based storage inherently unstable and unreliable.

Google Colab, while quick and easy to set up, is not designed for enterprise-level data handling. It struggles when accessing data stored outside its local disk (which in turn is limited to only around 250GB total)—even from Google Cloud Storage (GCS)—and cannot fully capitalize on streaming capabilities. Additionally, Colab runtimes are prone to unexpected termination without proper logging, which makes debugging multi-hour training jobs especially difficult.

To overcome these limitations, replacements were found in the form of Vertex AI and Google Cloud Services (GCS).

6.2 Transition to VertexAI and GCS

6.2.1 VertexAI

Vertex AI is significantly more complex to initiate compared to Colab, but once understood, training jobs can be set up and executed with high efficiency. It offers bandwidth, resources, and speeds far exceeding those of Colab. Particularly appealing is the access to powerful A100 GPU's as well as customizable local disc space - allowing for direct download of training data of any size into local memory, significantly improving data loading. As part of the Google Cloud ecosystem, Vertex AI provides seamless integration with GCS, reducing .tar file download times from over an hour to just seconds. Its support for detailed logging and long, stable runtimes is what ultimately enabled this research to move past the dataset iteration phase. Although individual accounts only offer access to the preemptible version of these resources - meaning Google is entitled to terminating your session whenever they see fit, this did not occur in this research, and on the odd possible chance that it does - one can restart the session from the last saved checkpoint. Overall Vertex AI is an enormous step up from Google Colab, and as far as large data processing is concerned should always be used over Google Colab - unless rapid setup is paramount.

6.2.2 Google Cloud Services (GCS)

GCS is part of the same “family” and offers a similarly robust capability for large-scale data handling. Support for tools like fsspec, allows for extremely fast data streaming to Vertex AI, there is no risk of timeout, and failed or erroneous client requests are clearly logged and handled without

interrupting the training code. Uploading to GCS is also multi-threadable meaning we can cut down on upload times too. Compared to Google Drive, GCS buckets emerge as a substantially more stable and powerful alternative.

6.2.3 Step by step rundown

To start off with we download the xBD dataset, and select the data subset we want to train on. This data is stored locally, and Universal Preprocessing steps are performed using code ran by Visual Studio Code. Having now the tiled, oversampled, normalized, etc. data we can upload it to our Bucket using Gsutil. From there we continue with our cloud-stored data, launching the training Jon in VertexAI, streaming the .tar libraries using fspec, training and saving the outputted model checkpoints in GCS. Below (*Figure 6*) I present a diagram of this process.

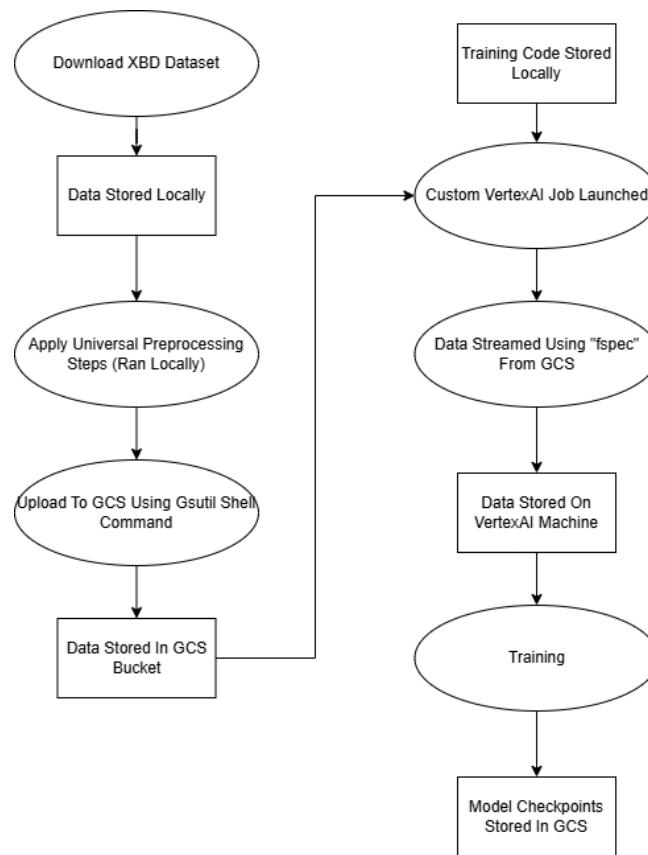


Figure 6 - Diagram of improved data pipeline

6.2.4 Resulting improvements

On the whole, implementing this workflow I was able to prepare new iterations in a matter of a few hours (data preprocessing, upload and job launching) and training times were cut in half (it is also worth noting that close to 50% of the new training time was spent calculating scores such as f1, accuracy recall etc, if one were to skip these calculations, a model could be trained at triple the speed of that shown by Vu ¹⁶). It is also significantly more reliable, no unexpected timeouts occurred and the training was capable of running overnight without human oversight.

6.3 Reverting to Colab/Drive for testing

Since testing is performed on a single MAXAR image pair, it is now viable to switch pipelines to one reminiscent of Nam's. The amount of data is fairly limited which lets Google Drive manage, and calculation speed is not of paramount importance, which allows us to work on Google Colab for running the testing code, and Google Drive for storage of the few files we need. In the case of this research this was primarily done to circumvent GCS billing, since the extensive tests (covered in Section 5) exhausted much of the available free credit.

7. Experiment Setup

The testing phase of this project is centered around evaluating the hybrid collapse detection pipeline on unseen real-world data, specifically, the high-resolution satellite images of Antakya—a region severely affected by the 2023 Turkey–Syria earthquake

The testing pipeline proceeds through the following stages, as depicted in *Figure 7*:

1. Data Preparation & Transfer

The raw MAXAR satellite imagery and HOTOSM geoJSON files for the Antakya region are downloaded locally and processed using the universal preprocessing pipeline (normalization, tiling, and image stacking). The resulting tiles are saved as .npy files.

These preprocessed image arrays are then uploaded to Google Drive for access via Colab.

2. Model Deployment in Google Colab

The UNet++ model—previously trained and saved as a checkpoint to Google Cloud Storage—is loaded into a Colab notebook. Input tiles are retrieved from Drive and passed through the model to produce an output masks, specifically a collapse probability map (probability_map2.png). The map is populated with values between 0 and 1 for each pixel, denoting its confidence in predicting a collapsed structure.

3. Shadow-Based Inference

Parallel to the neural network pipeline, the shadow algorithm is executed locally , generating its own collapse probability map (probability_map1.png) based the properties described in section 5.3.

4. Map Upload & Ensembling

The results of both UNet++ and the Shadow Model are uploaded to Drive and accessed again in Colab. The two maps are ensembled using soft-voting, where the final probability of collapse for each pixel is the average of both models' outputs. In cases where the shadow-based method is inconclusive (due to occlusion or lighting conditions), the ensemble falls back to using only the UNet++ prediction.

5. Thresholding & Postprocessing

The ensembled probability map (final_probability_map.png) is then thresholded to produce a binary collapse mask. This map is compared against the HOTOSM ground-truth annotations to evaluate performance. Threshold selection is empirically driven—optimizing for metrics

such as F1-score and Intersection-over-Union (IoU) on a validation subset if available.

6. **Validation**

Quantitative validation is conducted by overlaying the predicted masks with HOTOSM-derived labels. Metrics such as Precision, Recall, F1-score, and IoU are computed, and qualitative evaluation is also carried out by visual inspection in QGIS to understand prediction strengths and failure cases (e.g., shadow confusion in clustered buildings or occlusions).

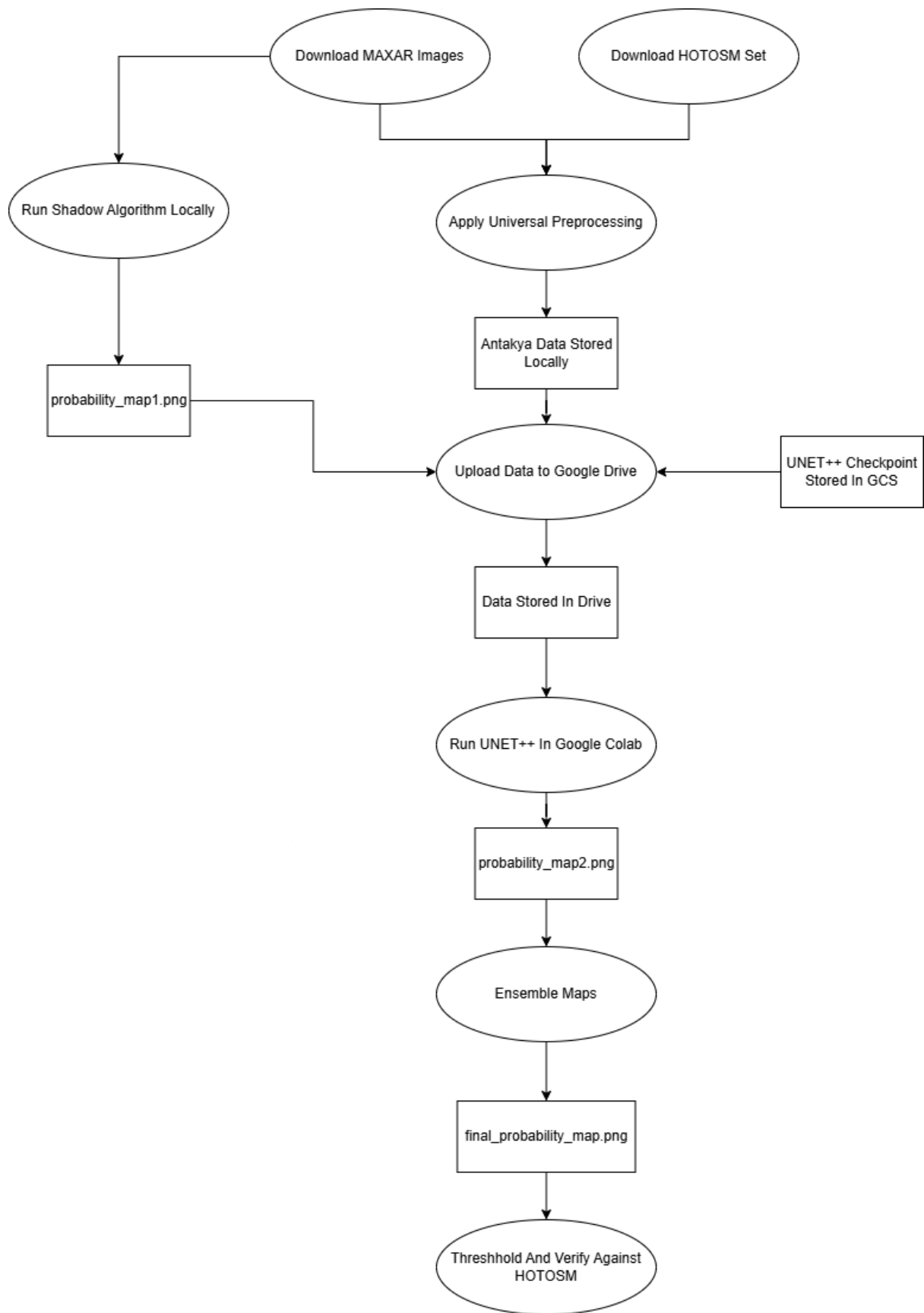


Figure 7 - Diagram of experiment flow

8. Results

This section evaluates the performance of the shadow-based method, the UNet++ model, and the hybrid ensemble across a series of threshold values (0.1–0.9). Since model outputs a pixel-level probability map, this is easily thresholded to produce binary collapse predictions. Performance is assessed using Accuracy, Precision, Recall, and F1 Score, calculated against HOTOSM-derived ground truth labels for the city of Antakya. Before going in depth into each separate algorithm, here are some summatory graph of the F1 Scores, Accuracy, Precision and Recall at each threshold (full data available in Appendix).

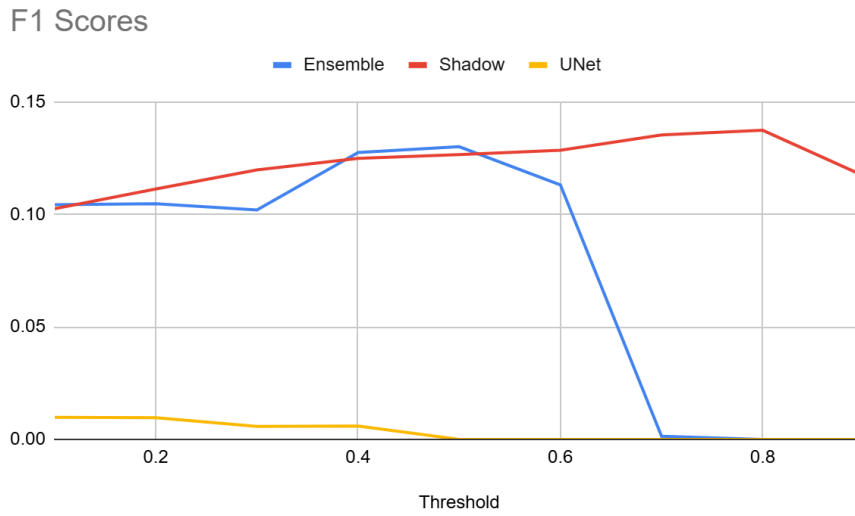


Figure 8 - F1 score graph

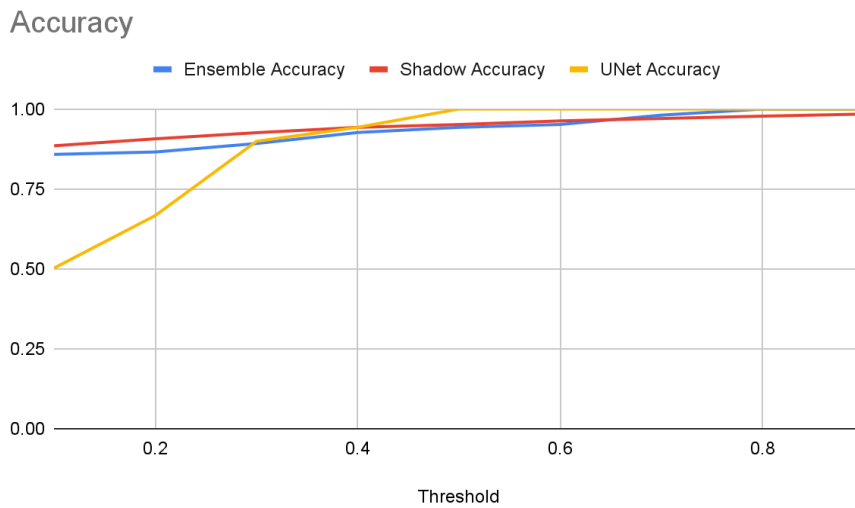


Figure 9 - Accuracy graph

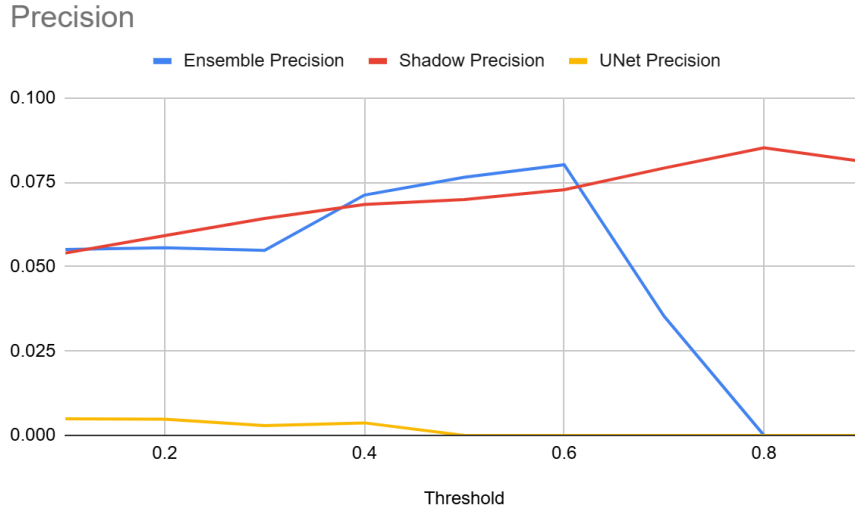


Figure 10 - Precision graph

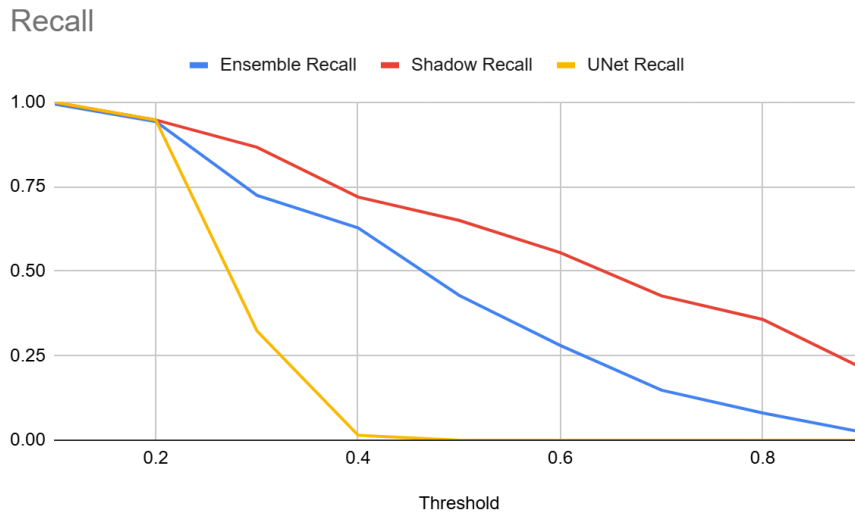


Figure 11 - Recall graph

8.1 Shadow-Based Performance

The Shadow Algorithm performed relatively well. Boasting a steadily increasing F1 score, which peaks around the 0.8 threshold at a value of 0.1374. The accuracy is very high (>90%) while precision steadily increases from around 0.05 to just over 0.08. Lastly recall shows a consistent drop from 1.0 to a minimum of 0.2113. These numbers, especially precision and accuracy are heavily impacted by the unbalanced nature of the task. Although correctly “catching” some of the collapses, each missed collapse and falsely flagged building is very costly.

8.2 UNet++ Performance

UNet++ showed significant limitations in generalizing to the Antakya dataset. At thresholds from 0.9 to 0.6, it predicted no collapsed pixels at all—leading to zero Precision, Recall, and F1. At

threshold 0.3 and below, minimal activity appears: at threshold 0.1, UNet++ achieves its best F1 score of 0.0098.

These results fall significantly short of those expected and exemplified by Nam Vu. This can be attributed to the significant difficulties in the very training of the algorithm. The unsolved issues of class imbalance and non-determinism in training seem to have produced a model that borders on unusable, even in a test setting.

8.3 Ensemble Performance

As we have seen the previous two sections, there is a vast divide between the Shadow and UNet performance. One achieves consistent, promising although slightly low results, while the other fails to come up with anything valuable. As a result the ensemble overall decreased performance from the pure shadow approach. Dropping scores pretty much across the board.

Despite this however there are some points where we can infer the possible benefits of this approach (assuming our UNet were working). For instance, at lower thresholds, the ensemble demonstrates relatively high recall, suggesting that it successfully aggregates predictions to capture more true positives. Additionally, the brief peak in F1 score and precision around the 0.4–0.5 threshold indicates that, under certain conditions, combining models can offer a balanced trade-off between sensitivity and specificity. This highlights the potential of ensemble methods to harness complementary strengths from diverse models—provided that each individual component contributes meaningfully. With a more reliable or better-trained UNet, the ensemble could potentially improve overall robustness and mitigate the limitations seen in single-model approaches. This points toward future work where optimizing the individual models before ensembling could lead to more consistently superior results.

8.4 Summary of Insights

Observation	Implication
Shadow model has steadily increasing precision but declining recall	Good balance at higher thresholds; suited for precision-critical tasks, though may miss collapses
Shadow achieves highest and most stable F1 performance	Most reliable among the models despite class imbalance
UNet++ fails to produce meaningful predictions	Model is poorly trained and unable to generalize; unusable in current state
Ensemble underperforms compared to Shadow alone	Weak UNet++ component drags down combined model performance
Accuracy stays high across methods	Dominance of background pixels masks class imbalance; F1 is more meaningful
Ensemble shows a brief F1 and precision peak around threshold 0.4–0.5	With stronger components, ensemble could balance recall and precision effectively

Table 3 - Table of insights

These results suggest that, in real-world deployment scenarios, the ensemble approach has the potential to offer the most balanced and operationally useful performance—particularly in time-sensitive, high-stakes contexts such as post-earthquake response. While further refinement of the UNet++ model is likely necessary to realize its full potential, the current framework already demonstrates that even modest individual model performance can be meaningfully enhanced through physically interpretable ensembling. This highlights the value of combining models with complementary strengths, especially when reliability and robustness are critical.

9. Discussion

This research aimed to critically evaluate and extend the collapse detection framework developed by Nam Vu, specifically targeting its generalization to real-world seismic events. While Vu reported promising performance metrics on the xBD dataset using UNet++, this thesis finds that these results are not easily replicable or generalizable without substantial architectural, data, and loss formulation refinements.

9.1 Model Performance in Context

Despite rigorous tuning and multiple architectural iterations, the UNet++ model alone failed to produce operationally useful results on the Antakya test set. Its behavior aligned with known issues in segmentation under severe class imbalance—predicting only background unless exposed to finely tuned weights and narrowly curated data. This confirms concerns raised in Vu’s own results, where even modest domain shifts led to performance degradation.

In contrast, the shadow-based algorithm—while simplistic—performed more consistently. With substantial metrics across the board and reasonable results on all thresholds (each with their own tradeoffs) It proved that that physically interpretable cues (like shadow geometry) can act as a stabilizing mechanism when black-box models falter.

There is concern however regarding the fixed building height (10m) across all shadow calculations. It introduces systemic noise and mid-rise buildings with intact structure may cast long shadows, appearing collapsed, while small buildings may appear undamaged due to minimal shadow loss. These false cues highlight the need for either building-specific metadata or further ensembling with Li-dar technology.

9.2 Ensemble Synergy and Limitations

The ensemble strategy effectively combined both approaches, although limited by the failure of UNet. Overall there should be little concern as to its effectiveness, in a worst-case scenario one can revert to choosing a single algorithm, however this should not be necessary. The ensemble itself does not have any inherent flaws, there are of course limitations to be considered - namely it will only be as good as its subcomponents, if one underperforms, the overall scored might be “dragged” down by the other, as we have seen in this paper. Outside of the issues of the modules themselves the ensemble seems to provide a good melting-pot of sorts for algorithms to strengthen each others weaknesses

9.3 Broader Challenges and Future Work

With all the successes and failures considered, let's now take a look at the deeper issues persisting across all model variants:

- Dataset generalization remains limited. xBD contains mixed-disaster imagery with only one major earthquake event. This undermines the transferability of models trained solely on its subsets.
- Lighting inconsistency in satellite images affects both learned features and shadow geometry, reducing confidence in predictions based on visual change.
- UNet++ instability was a consistent problem. Identical hyperparameters often produced diverging results.
- Building clusters produce previously unexplored shadow behaviours effectively “tricking” our shadow algorithm

Future work could address these with:

- Training on truly seismic-only datasets (e.g., curated collections from MAXAR or Multi-Scale Earthquake Damaged Buildings Feature Set (MEDBFS))
- Introducing metadata-driven conditional logic into the ensemble (e.g., weighting models differently per lighting condition)
- Augmenting the shadow module with object height priors from building registries or lidar-derived DEMs.
- Usage of seeding or similar deterministic methods in UNet++ application.
- Accounting for overlapping building shadow areas by correcting ratio for nearby building density.
- Curated order of images shown during training

10. Conclusions

This thesis presents a hybrid framework for rapid earthquake-induced building collapse detection using a combination of UNet++ semantic segmentation and shadow-based geometric analysis. Trained and tested on real satellite imagery from the 2023 Turkey–Syria earthquake.

Key contributions include:

- Vast Unet++ experimentation that explores the effects of class balancing, multi-class redefinition, and weighted loss functions
- A physics-driven shadow inference module that estimates structural loss without training

- A soft-voting ensemble that harmonizes learned predictions with shadow-derived validations to reduce false positives

Results show that while the UNet++ model underperforms in isolation, the ensemble strategy helps restore predictive value—especially under conditions where one model alone fails. This supports the central hypothesis that physically grounded methods can act as essential complements to deep learning in high-stakes, imbalanced, and noisy environments.

Ultimately, although impressive results were not achieved due to shadow behaviour and a troublesome UNet training process, significant work has been done to provide a much improved and detailed starting point for future research. Cutting down on training times, eliminating dead ends in the UNet specification, as well as uncovering critical behaviours in dense urban environment shadow interactions.

11. Appendix

Phase	Dataset	Sampling Strategy	Loss Function	Weights*1	Epochs	Result Quality	Notes	Metric Summary
Baseline Trials	Tier 1+3	Pos 5x	BCE+Dice	-	2	Collapsed	Initial experiment with basic oversampling. Performance nearly null.	Collapsed to one class. IoU: 0.002
Baseline Trials	Tier 1+3	Pos 10x	BCE+Dice	-	7	Collapsed	Double positive oversampling. No meaningful improvement observed.	Collapsed to one class. IoU: 0.002
Baseline Trials	Tier 1+3	Pos 10x / Neg 0.5x	Focal+Dice	-	15	Partial	Added negative undersampling and used Focal+Dice loss. Slight IoU bump.	Partial success. IoU: 0.134, modest recall
Dataset Shift	Tier 1	Pos 10x / Neg 0.5x	CE+Dice	-	15	Collapsed	Switched to Tier 1 dataset only. Model predicts only background.	Collapsed to one class. IoU: 0.045
Dataset Shift	Challenge	-	CE	-	15	Collapsed	First run on Challenge dataset. IoU improved, but still defaulting to background.	Collapsed to one class. IoU: 0.266

Weight Tuning	Challenge	-	CE	[0.0098, 0.1963, 1.0]	15	Success	Added data-driven weights. Model began to distinguish classes successfully.	Balanced output. Damage F1: 0.35, IoU: 0.15
Weight Tuning	Challenge	-	CE	[0.0098, 0.3, 1.0]	3	Collapsed	Increased No Damage weight too much. Model ignored other classes.	Collapsed to one class. IoU: 0.049
Weight Tuning	Challenge	-	CE	[0.0098, 0.25, 1.0]	4	Overfit	Reduced No Damage weight. Some improvement but instability appeared.	High background F1 (0.73), signs of overfitting
Loss Function Sweep	Challenge	-	CE+Dice	[0.0098, 0.1963, 1.0]	3	Collapsed	Reintroduced Dice loss. Model again leaned heavily on No Damage prediction.	Collapsed to one class. IoU: 0.049
Loss Function Sweep	Challenge	-	CE	[0.0098, 0.1963, 1.0]	5	Collapsed	Overweighted Damage class. Resulted in mode collapse to Damage.	Collapsed to one class. IoU: 0.038
Baseline Reversion	Challenge	-	CE	[0.0098, 0.1963, 1.0]	15	Collapsed	Reverted to original weights. Damage class dominates all predictions.	Collapsed to one class. IoU: 0.009

Table 4 - Full table of all loss function and data subset iteration

Threshold	Model	Accuracy	Precision	Recall	F1 Score
0.90	Shadow	0.9842	0.0811	0.2113	0.1171
	UNet	1.0000	0.0000	0.0000	0.0000
	Ensemble	1.0000	0.0000	0.0000	0.0000
0.80	Shadow	0.9777	0.0852	0.3571	0.1374
	UNet	1.0000	0.0000	0.0000	0.0000
	Ensemble	1.0000	0.0000	0.0000	0.0000
0.70	Shadow	0.9704	0.0792	0.4649	0.1353
	UNet	1.0000	0.0000	0.0000	0.0000

0.60	Ensemble	0.9809	0.0353	0.0007	0.0014
	Shadow	0.9629	0.0728	0.5495	0.1285
	UNet	1.0000	0.0000	0.0000	0.0000
0.50	Ensemble	0.9517	0.0802	0.1819	0.1131
	Shadow	0.9515	0.0699	0.6501	0.1266
	UNet	1.0000	0.0000	0.0000	0.0000
0.40	Ensemble	0.9427	0.0765	0.4283	0.1301
	Shadow	0.9427	0.0684	0.7193	0.1249
	UNet	0.9428	0.0037	0.0146	0.0059
0.30	Ensemble	0.9266	0.0712	0.6278	0.1275
	Shadow	0.9260	0.0643	0.8666	0.1198
	UNet	0.8992	0.0029	0.3231	0.0058
0.20	Ensemble	0.8921	0.0548	0.7241	0.1020
	Shadow	0.9068	0.0592	0.9471	0.1113
	UNet	0.6669	0.0048	0.9471	0.0096
0.10	Ensemble	0.8656	0.0556	0.9425	0.1047
	Shadow	0.8850	0.0540	0.9990	0.1025
	UNet	0.5025	0.0049	1.0000	0.0098
	Ensemble	0.8581	0.0551	0.9942	0.1043

Table 5 - Full table of results per model and threshold

12. References

1. Spence, R., So, E., & Scawthorn, C. (2020). Human casualties in earthquakes: Progress in modelling and mitigation. *Bulletin of Earthquake Engineering*, 18(3), 983–1021. <https://doi.org/10.1007/s10346-020-01512-5>
2. Guidoboni, E., & Comastri, A. (2009). The large earthquake of 8 August 1303 in Crete: Seismic scenario and tsunami in the Eastern Mediterranean area. *Quaternary International*, 196(1–2), 90–105. <https://doi.org/10.1016/j.quaint.2009.05.005>

3. Mileti, D. S. (2004). Disasters by Design: A Reassessment of Natural Hazards in the United States. *Landscape and Urban Planning*, 69(2–3), 195–196.
<https://doi.org/10.1016/j.landurbplan.2004.11.005>
4. Lin, A., & Ren, Z. (2013). Tectonic deformation and paleoseismicity along the Longmen Shan fault zone before and after the 2008 Wenchuan earthquake, central China. *Journal of Asian Earth Sciences*, 74, 9–23. <https://doi.org/10.1016/j.jseaes.2013.02.027>
5. Liu, W., Wang, Y., Chen, S., & Yu, M. (2021). The impact of urbanization on earthquake vulnerability in China. *International Journal of Disaster Risk Science*, 12(3), 347–359.
<https://doi.org/10.1007/s13753-021-00385-z>
6. Geiß, C., Taubenböck, H., Wurm, M., & Lakes, T. (2015). Remote sensing-based characterization of settlement structures for assessing local earthquake risk. *Natural Hazards and Earth System Sciences*, 15(5), 931–945. <https://doi.org/10.5194/nhess-15-931-2015>
7. Debacker, M., Hubloue, I., Dhondt, E., & Knockaert, D. (2003). Medical interventions after earthquakes: Review of international literature. *European Journal of Emergency Medicine*, 10(4), 303–308. <https://doi.org/10.1007/s00068-003-1239-y>
8. Kim, H., Lee, J., Kim, Y., & Kim, S. (2024). Burn injuries from structural collapse incidents: A retrospective study. *Burns*, in press. <https://doi.org/10.1016/j.burns.2024.03.001>
9. Bartels, S. A., & VanRooyen, M. J. (2011). Medical complications associated with earthquakes. *The Lancet*, 379(9817), 748–757.
[https://doi.org/10.1016/S0140-6736\(11\)60887-8](https://doi.org/10.1016/S0140-6736(11)60887-8)
10. Kovács, G., & Tatham, P. (2009). Responding to disruptions in the supply network: From source to aid recipients. *International Journal of Physical Distribution & Logistics Management*, 39(2), 102–120. <https://doi.org/10.1108/09600030710734820>
11. Masudin, I., & Saputra, A. R. (2019). An application of benchmarking for humanitarian logistics: An analytical approach. *Benchmarking: An International Journal*, 27(1), 193–211.
<https://doi.org/10.1108/BIJ-04-2019-0165>
12. Ma, X., Yu, M., Gong, L., & Song, X. (2024). Early detection of trapped victims after earthquakes: A rapid triage method. *BMC Emergency Medicine*, 24, Article 49.
<https://doi.org/10.1186/s12873-024-00949-4>
13. Noji, E. K. (2011). Earthquakes. *Disaster Medicine and Public Health Preparedness*, 5(1), 56–61. <https://doi.org/10.1001/dmp.2011.5>
14. Zhou, C., Zhang, J., Wang, X., & Zhang, Y. (2022). A review of convolutional neural networks in medical image analysis. *International Journal of Computational Biology and Drug Design*, 15(1), 19–45. <https://doi.org/10.1142/S1793545822500316>
15. Jaiswal, A., Gianchandani, N., Singh, D., Kumar, V., & Kaur, M. (2021). Classification of COVID-19 patients from chest CT images using multi-objective differential evolution-based

- convolutional neural networks. *Computers in Biology and Medicine*, 134, 105093. <https://doi.org/10.1016/j.compbimed.2021.105093>
16. Vu, N. H. (2024). *UNET++ Convolution Neural Network for Building Damage Assessment: A Case Study of the 2023 Turkey–Syria Earthquakes* [Bachelor thesis, Vrije Universiteit Amsterdam].
 17. Matin, S. S., & Pradhan, B. (2021). Earthquake-induced building-damage mapping using explainable AI (XAI). *Sensors*, 21(13), 4489. <https://doi.org/10.3390/s21134489>
 18. Tong, X., Liu, S., Li, T., & Zhan, W. (2013). A building damage assessment method integrating high-resolution imagery and ground information. *Journal of Applied Remote Sensing*, 7(1), 073532. <https://doi.org/10.1117/1.JRS.7.073532>
 19. Raju, P. L. N., & Lakshmi, T. V. (n.d.). Shadow analysis technique for extraction of building height using high resolution satellite single image and accuracy assessment. *ISRS Conference Paper*.
 20. Zhou, Z., Siddiquee, M. M. R., Tajbakhsh, N., & Liang, J. (2018). UNet++: A Nested U-Net Architecture for Medical Image Segmentation. *ResearchGate*. <https://www.researchgate.net/publication/324574642>
 21. Matin, S. S., & Pradhan, B. (2021). Challenges and limitations of earthquake-induced building damage mapping techniques using remote sensing images—A systematic review. *Geocarto International*, 37(21), 6186–6212. <https://doi.org/10.1080/10106049.2021.1933213>
 22. Ahmad, K., Maabreh, M., Ghaly, M., Khan, K., Qadir, J., & Al-Fuqaha, A. (2020). Developing future human-centered smart cities: Critical analysis of smart city security, interpretability, and ethical challenges. *ResearchGate*. <https://researchgate.net/publication/347398615>
 23. Nex, F., Duarte, D., Steenbeek, A., & Kerle, N. (2019). Towards real-time building damage mapping with low-cost UAV solutions. *Remote Sensing*, 11(3), 287. <https://doi.org/10.3390/rs11030287>
 24. Maxar Technologies. (2023). Turkey earthquake 2023: Open data. Retrieved from <https://www.maxar.com/open-data/turkey-earthquake-2023>
 25. Humanitarian Open Street Map Team. (2023). HOTOSM Turkey destroyed buildings (2023 Kahramanmaras Earthquake). Retrieved from https://data.humdata.org/dataset/hotosm_tur_destroyed_buildings
 26. GitHub link to all code: <https://github.com/RAGNAROS-HS/Thesis2025.git>
 27. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. arXiv. <https://arxiv.org/abs/1505.04597>

28. xView2. (n.d.). xView2: Building damage assessment. Retrieved from <https://xview2.org/dataset>